Nickolas-Adrian Papamichail

QUANTUM SOFTWARE | CIRCUIT OPTIMIZATION | PYTHON AND C++

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LinkedIn: Nickolas-Adrian Papamichail | English: professional working proficiency

PROFILE

Informatics and Telecommunications undergraduate focused on quantum software and computational modeling. Thesis research combines quantum circuit optimization, episodic memory and controlled benchmark comparisons using Python and Qiskit. Seeking quantum software internships or junior roles in quantum applications and hybrid computing.

TECHNICAL SKILLS

Quantum computing Qiskit, IBM Quantum Composer, quantum circuits, phase rotations
Programming and tools Python, C++, C; Git, Linux, CMake, VS Code
Scientific computing NumPy, SciPy; Hamiltonian modeling and eigenvalue analysis
Evaluation Circuit gate counts and depth, memory ablation, regression checks

RESEARCH

Cognitive Optimizer | Bachelor thesis experiment · Python, Qiskit

- Implemented a classical optimizer for quantum circuits that merges phase rotations and reuses an episodic memory of the transformation rule.
- Compared memory-enabled and memory-disabled execution on a 12-qubit VQE circuit and a 10-qubit synthetic circuit; tracked gate count, depth and CX count.
- Recorded 107 memory recalls out of 108 rule applications on the VQE benchmark and 33 out of 34 on the synthetic benchmark; consistency and regression checks passed.

SELECTED PROJECTS

Graphene Lattice Simulation | Hamiltonian modeling

- Modeled a graphene lattice and analyzed its eigenvalues to investigate electronic states.

Quantum Collagen Optimization | Exploratory modeling · IBM Quantum Composer

- Created a quantum circuit implementation for an exploratory collagen optimization project.

BioMapX / Q-Chaperone | C++ simulation architecture · in development

- Designed modular simulation state and computational workloads for molecular modeling, with hybrid quantum-classical scheduling as a planned extension.

EDUCATION

University of Ioannina | BSc in Informatics and Telecommunications
GPA 8.75/10 | Expected graduation September 2026
Thesis Beyond Neural: A Quantum Approach to Artificial Cognition

SELECTED TRAINING

Integrating Quantum and High-Performance Computing | IBM, May 2026
IBM Quantum Machine Learning | IBM, March 2026
Elements of Supercomputing | Kajaani University of Applied Sciences, 2026

AWARDS

Award of Distinction, Mathematical Riddles Competition | 2021
1st Online Student Mathematics Festival